

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I J. Manoj (19957143) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

J. Manoj
Signature of the student:

65% 

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20% ✓	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20% ✓	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25% ✓	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20% ✓	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15% ✓	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

	A	B	C	D	E			
	A	B	C	D	E	F	G	H
Q ₁	2	1	2	1.5	1.5		8	
Q ₂	1	1	1.5	1.5	1		6	
Q ₃	1	1.5	1	2	1		6.5	32.5
Q ₄	1	1	1	2	1		6	
Q ₅	1	1	1.5	1.5	1		6	

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I __j.manoj(192572143)__ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

j.manoj

Annexure A

ANALYTICAL PROBLEM SOLVING — SOLUTIONS

QUESTION 1:

A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

Solution:

Given block: 192.168.20.0/24, divided into Lab A (/26), Lab B (/27), Lab C (/28), allocated sequentially from the start of the block.

Lab A — /26 (Subnet Mask 255.255.255.192, Block size 64)

Network Address	192.168.20.0
Valid Host Range	192.168.20.1 – 192.168.20.62
Broadcast Address	192.168.20.63
Usable Hosts	$2^6 - 2 = 62$

Lab B — /27 (Subnet Mask 255.255.255.224, Block size 32)

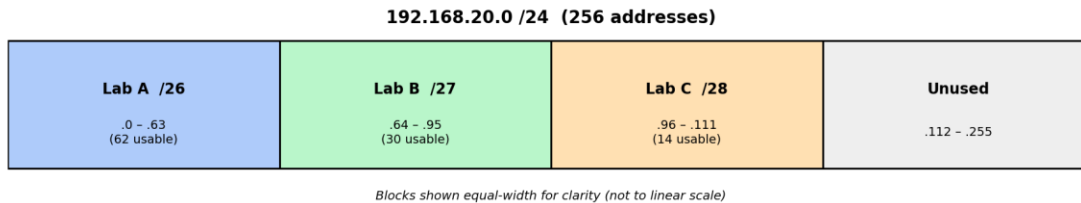
Network Address	192.168.20.64
Valid Host Range	192.168.20.65 – 192.168.20.94
Broadcast Address	192.168.20.95
Usable Hosts	$2^5 - 2 = 30$

Lab C — /28 (Subnet Mask 255.255.255.240, Block size 16)

Network Address	192.168.20.96
Valid Host Range	192.168.20.97 – 192.168.20.110
Broadcast Address	192.168.20.111
Usable Hosts	$2^4 - 2 = 14$

Interpretation: The three labs fit within 192.168.20.0/24 without overlap (Lab A uses .0–.63, Lab B uses .64–.95, Lab C uses .96–.111), leaving 192.168.20.112 – 192.168.20.255 unused for future expansion.

Q1: Subnetting 192.168.20.0/24 into Lab A (/26), Lab B (/27), Lab C (/28)



QUESTION 2:

A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

Solution:

System 1: 192.168.1.130/26 | System 2: 192.168.1.190/26 (Subnet Mask 255.255.255.192, Block size 64). The /26 blocks along the fourth octet are: 0–63, 64–127, 128–191, 192–255.

System 1 — 192.168.1.130

Subnet block	128 – 191
Network Address	192.168.1.128
Broadcast Address	192.168.1.191
Valid Host Range	192.168.1.129 – 192.168.1.190
Falls within subnet?	Yes — 130 lies between 129 and 190

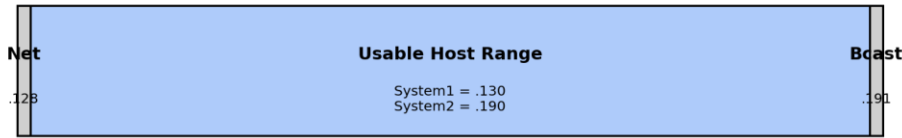
System 2 — 192.168.1.190

Subnet block	128 – 191
Network Address	192.168.1.128
Broadcast Address	192.168.1.191
Valid Host Range	192.168.1.129 – 192.168.1.190
Falls within subnet?	Yes — 190 is the last usable host address in the range

Both System 1 and System 2 belong to the SAME subnet, 192.168.1.128/26, since both addresses fall between the network address (192.168.1.128) and broadcast address (192.168.1.191). Usable host addresses in this subnet: $2^6 - 2 = 62$.

Q2: Both systems fall in the same /26 subnet 192.168.1.128/26

192.168.1.x (subnet block 128-191)



QUESTION 3:

An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR $\rightarrow$ /25, Sales $\rightarrow$ /26, IT $\rightarrow$ /27, and Admin $\rightarrow$ /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

Solution

Given block: 172.16.0.0/16, VLSM subnetting: HR → /25, Sales → /26, IT → /27, Admin → /28, allocated sequentially, largest subnet first.

HR — /25 (Block size 128)

Network Address	172.16.0.0
Valid Host Range	172.16.0.1 – 172.16.0.126
Broadcast Address	172.16.0.127
Usable Hosts	$2^7 - 2 = 126$

Sales — /26 (Block size 64)

Network Address	172.16.0.128
Valid Host Range	172.16.0.129 – 172.16.0.190
Broadcast Address	172.16.0.191
Usable Hosts	$2^6 - 2 = 62$

IT — /27 (Block size 32)

Network Address	172.16.0.192
Valid Host Range	172.16.0.193 – 172.16.0.222
Broadcast Address	172.16.0.223

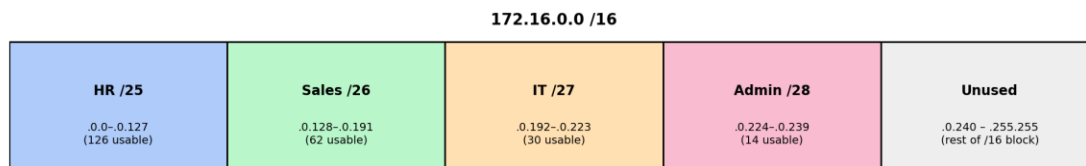
Usable Hosts	$2^5 - 2 = 30$
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Admin — /28 (Block size 16)

Network Address	172.16.0.224
Valid Host Range	172.16.0.225 – 172.16.0.238
Broadcast Address	172.16.0.239
Usable Hosts	$2^4 - 2 = 14$

Remaining unused IP address range within 172.16.0.0/16: 172.16.0.240 – 172.16.255.255 (available for future departments or future expansion).

Q3: VLSM Subnetting of 172.16.0.0/16



Blocks shown equal-width for clarity (not to linear scale)

QUESTION 4:

A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

Solution

Given: 172.16.10.0/24 subnetted using /27 (Subnet Mask 255.255.255.224, Block size 32).

Total subnets created	$2^{(27-24)} = 2^3 = 8$ subnets
Usable hosts per subnet	$2^{(32-27)} - 2 = 2^5 - 2 = 30$

Subnet containing 172.16.10.78

The /27 blocks in the last octet are: 0–31, 32–63, 64–95, 96–127, 128–159, 160–191, 192–223, 224–255. The address .78 falls in the block 64–95.

Network Address	172.16.10.64
Valid Host Range	172.16.10.65 – 172.16.10.94
Broadcast Address	172.16.10.95

Does 172.16.10.95 fall in the same subnet?

Yes — 172.16.10.95 lies within the 172.16.10.64/27 block, but it is the BROADCAST address of that subnet, not a usable host address. So it belongs to the same subnet range, but cannot be assigned to a host.

Q4: 172.16.10.0/24 divided into 8 subnets of /27

172.16.10.0 /24 → 8 × /27 subnets, 30 usable hosts each

Subnet 1 /27	Subnet 2 /27	Subnet 3 /27	Subnet 4 /27	Subnet 5 /27	Subnet 6 /27	Subnet 7 /27	Subnet 8 /27
.0-.31	.32-.63	.64-.95 (.78 here)	.96-.127	.128-.159	.160-.191	.192-.223	.224-.255

172.16.10.78 and 172.16.10.95 both fall in Subnet 3 (.64-.95); .95 is its broadcast address

QUESTION 5:

A network uses the Internet Protocol version 6 address `2001:0db8:0000:0000:0000:ff00:0042:8329`. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

Solution

Given IPv6 address: `2001:0db8:0000:0000:0000:ff00:0042:8329`

Compressed (shortest) form

Rule 1 — remove leading zeros in each group. Rule 2 — replace the longest run of consecutive all-zero groups with '::' (once only).

Compressed Address	<code>2001:db8::ff00:42:8329</code>
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Expanded (full) form

Re-expanding the compressed address restores every group to 4 hex digits:

Expanded Address	<code>2001:0db8:0000:0000:0000:ff00:0042:8329</code>
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Network prefix at /64

The first 64 bits (first four groups) form the network prefix; the remaining 64 bits are the interface ID.

Network Prefix	<code>2001:0db8:0000:0000::/64</code> (i.e., <code>2001:db8::/64</code>)
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Total possible host addresses in a /64

Host portion = $128 - 64 = 64$ bits.

Total Host Addresses	$2^{64} = 18,446,744,073,709,551,616$
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Q5: IPv6 Address Structure — 2001:0db8:0000:0000:0000:ff00:0042:8329 /64

